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Capstone Project Report

Explainable Chest X-Ray Report Generation with Built-in Hallucination Detection

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Abstract

Chest X-ray interpretation and radiology report generation are important components of clinical diagnosis, but the increasing workload and shortage of radiologists can result in delays in reporting. Recent advances in deep learning and vision-language models have enabled automated generation of radiology reports from chest X-ray images. However, existing report-generation systems primarily focus on improving linguistic and clinical generation metrics and generally lack mechanisms for verifying whether the generated findings are supported by the input image or consistent with available patient information. This limitation can lead to clinically unsafe hallucinations in automatically generated reports.

This work proposes an explainable chest X-ray report generation framework with a built-in dual hallucination detection mechanism. The proposed system first uses a DenseNet121-based multi-label classifier to identify 16 canonical chest X-ray findings and employs Grad-CAM to provide visual explanations for the predicted findings. A retrieval-augmented generation (RAG) module based on FAISS retrieves relevant prior cases and medical guidelines to provide additional contextual information. Patient history is also incorporated into the report-generation process. A Qwen2-VL-2B vision-language model then generates a radiology report using the X-ray image, retrieved information, and patient context. To improve the reliability of the generated report, a dual hallucination detection mechanism checks generated findings against both image-based classifier predictions and patient history. The first check identifies image-unsupported findings, while the second detects clinically inconsistent findings. The resulting system aims to combine automated report generation, visual explainability, contextual retrieval, and automated verification within a unified pipeline.

Keywords: Chest X-Ray, Radiology Report Generation, Explainable AI, Hallucination Detection, DenseNet121, Grad-CAM, Retrieval-Augmented Generation, Qwen2-VL, Medical Vision-Language Models

1 Introduction

Chest X-ray imaging is one of the most widely used diagnostic imaging modalities in clinical practice. Radiologists examine chest X-rays to identify abnormalities and subsequently prepare textual reports describing their observations and conclusions. With the increasing volume of medical imaging studies and the shortage of radiologists, automated systems capable of assisting with report generation have gained significant research attention. Automated radiology report generation has the potential to reduce reporting time and support clinical workflows by converting visual findings into structured clini-

cal narratives. However, reliability and clinical safety remain major challenges for these systems.

Recent research has explored knowledge-enhanced multimodal attention, self-supervised learning, Transformer-based architectures, contrastive learning, expert knowledge retrieval, cross-modal interaction, and reinforcement learning to improve automatic radiology report generation. These approaches have demonstrated improvements in visual representation, multimodal alignment, report fluency, and clinical content generation. However, their primary emphasis remains on improving the quality of generated reports rather than independently verifying whether the generated clinical claims are supported by the input X-ray.

A major concern in medical vision-language systems is hallucination, where a model generates information that is not supported by the available evidence. In the context of chest X-ray report generation, hallucinations may appear as findings that are absent from the X-ray image. In addition, generated findings may conflict with information contained in a patient's clinical history. Such errors can reduce the reliability of automated reporting systems and create potential risks if generated reports are used without adequate verification. The proposed work therefore considers two distinct hallucination categories: image-unsupported hallucinations and clinical inconsistency hallucinations.

Another limitation of automated medical reporting systems is their lack of interpretability. Black-box predictions may provide a diagnosis or generated statement without clearly indicating the visual evidence responsible for the prediction. For clinical applications, visual explanations can assist radiologists in understanding and verifying model predictions. In this work, Grad-CAM is incorporated with the image classification component to provide heatmaps highlighting regions associated with predicted findings.

To address these limitations, this work proposes an end-to-end explainable chest X-ray report generation framework. The system combines a DenseNet121-based multi-label finding classifier, Grad-CAM-based visual explanations, retrieval-augmented generation using FAISS, patient-history integration, Qwen2-VL-2B-based report generation, and a dual hallucination detection mechanism. The overall pipeline processes the X-ray image, identifies relevant findings and visual evidence, retrieves contextual information, generates a draft report, and finally verifies the generated findings before producing the final output.

2 Literature Review & Table

Automatic radiology report generation has evolved from conventional image-to-text approaches toward multimodal systems that incorporate medical knowledge, attention mechanisms, contrastive learning, retrieval, and large vision-language models. Recent studies

have primarily focused on improving the quality, relevance, and clinical content of automatically generated reports.

Xu et al. (2023) proposed *Radiology Report Generation via Structured Knowledge-Enhanced Multi-modal Attention and Contrastive Learning*, which combines structured medical knowledge with multimodal attention and dual-branch contrastive learning. Although knowledge fusion improves report generation, the approach does not provide a dedicated mechanism for determining whether generated clinical claims are supported by the input image [1].

Pan et al. (2024) introduced S3-Net, a self-supervised dual-stream network that combines CNN and Swin Transformer representations with a Fusion Alignment Module. The approach improves visual representation and visual-text alignment, but its primary focus remains report generation rather than independent verification of generated findings [2].

Yi et al. (2024) proposed the Two-Stage Global Enhanced Transformer (TSGET), which connects global image features with generated words and subsequently incorporates region-level visual information. While the model addresses global and regional visual representation, it primarily optimizes report-generation performance rather than detecting unsupported clinical statements [3].

Liu et al. (2024) proposed *Multi-Grained Radiology Report Generation with Sentence-Level Image-Language Contrastive Learning*. The method learns sentence-level topics and fine-grained image representations using contrastive learning and employs multiple decoders for report generation. The study identifies challenges involving rare findings, subtle abnormalities, uncertainty, and long or complex sentences [4].

Zhang et al. (2024) developed a Visual-Textual Cross-Modal Interaction Network to address the problem of abnormal regions occupying only a small portion of chest X-ray images. The model improves visual-textual interaction and abnormality representation but does not introduce a post-generation verification mechanism [5].

Bu et al. (2024) proposed EKAGen, which incorporates instance-level expert knowledge, discriminative attention, and self-distillation to improve disease-specific visual representation. Although the approach makes better use of expert knowledge and important image regions, it does not independently verify each generated clinical claim [6].

Gu et al. (2024) introduced an Automatic Medical Report Generation approach based on a Detector Attention Module and GPT-Based Word LSTM. The detector attention module provides lesion-location information while GPT-based semantic information contributes to report generation. However, the approach does not explicitly determine whether generated statements are clinically supported [7].

Hou et al. (2024) proposed Retrieval Enhanced Cross-modal Fusion for radiographic

report generation. Retrieval is used to provide relevant information during generation, demonstrating the potential of retrieval-based contextual enhancement. However, retrieval alone does not guarantee that the final generated report is faithful to the current X-ray image [8].

Liao et al. (2024) investigated Information Extraction to Normalize Training Data for Automatic Radiology Report Generation, using clinical entity and relation extraction to transform narrative reports into structured representations. The study also highlights limitations related to clinical context, uncertain observations, and generalization across different clinical sites [9].

Yi et al. (2025) proposed LHR-RFL, a linear hybrid-reward-based reinforced focal learning approach that gives greater importance to difficult training cases. The work focuses on improving report-generation performance on difficult samples but does not introduce a dedicated verification layer for hallucinated findings [10].

2.1 Literature Review Table

Table 1 summarizes the ten reviewed studies, their major approaches, contributions, and limitations relevant to the proposed work.

Table 1: Summary of the Reviewed Literature

Paper / Year	**Method**	**Key Contribution**	**Relevant Limitation**
Xu et al. (2023)	Knowledge-enhanced multimodal attention and contrastive learning	Combines structured medical knowledge with multimodal attention and dual-branch contrastive learning for radiology report generation.	Does not provide a dedicated mechanism for verifying whether generated clinical claims are supported by the input image.
Pan et al. (2024)	S3-Net; self-supervised dual-stream CNN and Swin Transformer	Uses dual-stream visual representations and a Fusion Alignment Module to improve visual-text alignment.	Focuses primarily on report generation and does not independently verify generated findings.

Paper / Year	Method	Key Contribution	Relevant Limitation
Yi et al. (2024)	Two-Stage Global Enhanced Transformer (TS-GET)	Combines global image features with generated words and incorporates region-level visual information.	Primarily optimizes report generation rather than detecting unsupported clinical statements.
Liu et al. (2024)	Multi-grained report generation with sentence-level image-language contrastive learning	Learns sentence-level topics and fine-grained image representations using contrastive learning.	Challenges remain for rare findings, subtle abnormalities, uncertainty, and complex reports.
Zhang et al. (2024)	Visual-Textual Cross-Modal Interaction Network	Improves visual-textual interaction for abnormal regions occupying relatively small image areas.	Does not introduce a dedicated post-generation verification mechanism.
Bu et al. (2024)	EKAGen; expert knowledge-aware report generation	Uses instance-level expert knowledge, discriminative attention, and self-distillation for disease-specific representation.	Does not independently verify each generated clinical claim.
Gu et al. (2024)	Detector Attention Module and GPT-Based Word LSTM	Combines lesion-location information from detector attention with GPT-based semantic information for report generation.	Does not explicitly determine whether generated statements are clinically supported.
Hou et al. (2024)	Retrieval Enhanced Cross-modal Fusion	Uses retrieval to provide relevant information during radiographic report generation.	Retrieved information alone does not guarantee faithfulness to the current X-ray image.

Paper / Year	Method	Key Contribution	Relevant Limitation
Liao et al. (2024)	Information extraction for training-data normalization	Uses clinical entity and relation extraction to transform narrative reports into structured representations.	Highlights limitations related to clinical context, uncertain observations, and cross-site generalization.
Yi et al. (2025)	LHR-RFL; hybrid-reward reinforced focal learning	Emphasizes difficult training cases to improve radiology report-generation performance.	Does not introduce a dedicated verification layer for hallucinated findings.

2.2 Research Gap

The reviewed literature demonstrates substantial progress in automatic chest X-ray report generation through improved visual representation, multimodal alignment, medical knowledge integration, retrieval, and training strategies. However, the reviewed approaches primarily concentrate on generation quality. According to the literature survey conducted for this work, the existing approaches do not provide a dedicated mechanism that simultaneously verifies generated findings against image evidence and patient clinical history.

This motivates the development of a dual hallucination detection mechanism capable of identifying both findings that are unsupported by the X-ray and findings that are inconsistent with patient history.

3 Problem Statement and Objectives

3.1 Problem Statement

Automated chest X-ray report generation systems have demonstrated the ability to convert medical images into textual radiology reports. However, generated reports may contain findings that are not supported by the input image or statements that are inconsistent with available patient information. Existing approaches also provide limited visual interpretability regarding the image regions responsible for predicted findings.

Therefore, there is a need for an explainable chest X-ray report generation framework

that can generate clinically meaningful reports while also providing visual evidence and explicitly verifying generated findings against both image-based evidence and patient history.

3.2 Objectives

The main objectives of the proposed work are:

1. To develop a DenseNet121-based multi-label classifier for identifying 16 canonical chest X-ray findings.
2. To provide visual explanations for predicted findings using Grad-CAM.
3. To incorporate relevant prior cases and medical guideline information using retrieval-augmented generation.
4. To integrate available patient history as additional clinical context during report generation and verification.
5. To generate automated radiology reports using the Qwen2-VL-2B vision-language model.
6. To develop a dual hallucination detection mechanism for identifying image-unsupported and clinically inconsistent findings.
7. To combine the generated report, predicted findings, hallucination flags, and Grad-CAM visualization into an explainable final output.

4 Existing System and Proposed System

4.1 Existing System

Existing automated chest X-ray report generation systems generally follow an image-to-text generation paradigm. A deep learning model extracts visual features from a chest X-ray and a language-generation component converts these features into a textual radiology report.

Recent systems have incorporated attention mechanisms, Transformer architectures, contrastive learning, medical knowledge, retrieval, and reinforcement learning to improve report quality. These methods have contributed to improvements in visual representation, image-text alignment, linguistic fluency, and clinical content generation.

However, report generation is generally treated as the final stage of the system. The generated report may not be independently checked against the image evidence or patient history. Consequently, unsupported clinical statements may remain in the generated report.

4.2 Limitations of Existing Systems

The major limitations considered in this work are:

- **Hallucinated findings:** A report-generation model may generate findings that are not supported by the input X-ray.
- **Limited verification:** Generated clinical claims are not necessarily checked against independent image-based predictions.
- **Limited patient-context integration:** Patient history may not be explicitly incorporated into report generation or verification.
- **Limited explainability:** Generated findings may not clearly indicate which image regions contributed to the underlying prediction.
- **Knowledge limitations:** A generation model may not have access to relevant prior cases or external clinical context during inference.

4.3 Proposed System

The proposed system addresses these limitations by integrating multiple components into a unified pipeline.

First, a DenseNet121-based multi-label classifier identifies 16 canonical clinical findings from the chest X-ray. Grad-CAM is then used to generate visual explanations for the predicted findings.

Second, a retrieval-augmented generation module based on FAISS retrieves relevant prior cases and medical guideline information. Patient history is incorporated as an additional clinical context.

Third, Qwen2-VL-2B receives the chest X-ray together with the retrieved information and patient context and generates the radiology report.

Finally, the generated report is passed through a dual hallucination detection mechanism. The first verification stage compares generated findings against image-supported findings, while the second checks the generated findings against available patient history.

4.4 Advantages of the Proposed System

The proposed framework provides the following capabilities:

- Multi-label chest X-ray finding classification.
- Visual explanation through Grad-CAM.
- Retrieval of relevant clinical knowledge and prior cases.
- Patient-history-aware report generation.
- Vision-language-based automated report generation.
- Image-supported hallucination detection.
- Clinical-consistency hallucination detection.
- An integrated final output containing report, findings, verification flags, and visual explanation.

5 Data Set Description

5.1 IU Chest X-Ray Dataset

The proposed system uses the IU Chest X-Ray (IU-CXR) dataset, associated with Indiana University and the Open-i biomedical image search system. The dataset contains chest X-ray images together with corresponding radiology reports, making it suitable for multimodal chest X-ray analysis and automatic radiology report generation.

The dataset used in this work contains 3,851 reports and 7,466 X-ray images. The image and report information is distributed across two principal CSV files, namely `indiana_projections.csv` and `indiana_reports.csv`. These files are linked through a common `uid`, which serves as the study identifier.

The `indiana_projections.csv` file contains 7,466 image records with three primary fields: `uid`, `filename`, and `projection`. The projection field identifies whether an image corresponds to a frontal or lateral view. Among the available images, 3,818 are frontal projections and 3,648 are lateral projections. Most studies contain a pair of frontal and lateral images.

The `indiana_reports.csv` file contains 3,851 study-level records and includes fields such as `uid`, `MeSH`, `Problems`, `image`, `indication`, `comparison`, `findings`, and `impression`. Among these fields, the `findings` field contains the detailed radiologist observations and

serves as the primary textual ground truth for the report-generation task. The `MeSH` and `Problems` fields provide structured clinical information used in the finding-label extraction process.

A total of 514 studies contain no findings text and were excluded from further processing, resulting in 3,337 usable reports. Missing values in the `indication` and `comparison` fields were retained and flagged rather than artificially imputed.

For the finding-classification component, the available clinical information is mapped into 16 canonical finding labels using multi-label binarization. Since a single chest X-ray may contain multiple findings simultaneously, the task is formulated as a multi-label classification problem rather than a mutually exclusive multi-class classification problem.

The resulting data is divided into 80% training, 10% validation, and 10% testing subsets. Patient-level splitting is used to ensure that images belonging to the same patient do not appear across different subsets, thereby reducing the risk of data leakage. The resulting test set contains 335 cases.

5.2 Dataset Statistics

Table 2: Summary of the IU Chest X-Ray Dataset

Property	**Value**
Dataset	IU Chest X-Ray
Total reports	3,851
Usable reports	3,337
Total X-ray images	7,466
Frontal images	3,818
Lateral images	3,648
Canonical findings	16
Training split	80%
Validation split	10%
Test split	10%
Test cases	335
Image-report linkage	UID

The combination of chest X-ray images, radiologist findings, structured clinical information, and contextual fields makes the IU-CXR dataset suitable for the proposed integrated framework involving finding classification, explainability, retrieval-augmented generation, report generation, and hallucination detection.

6 Proposed System Architecture

6.1 Overall Architecture

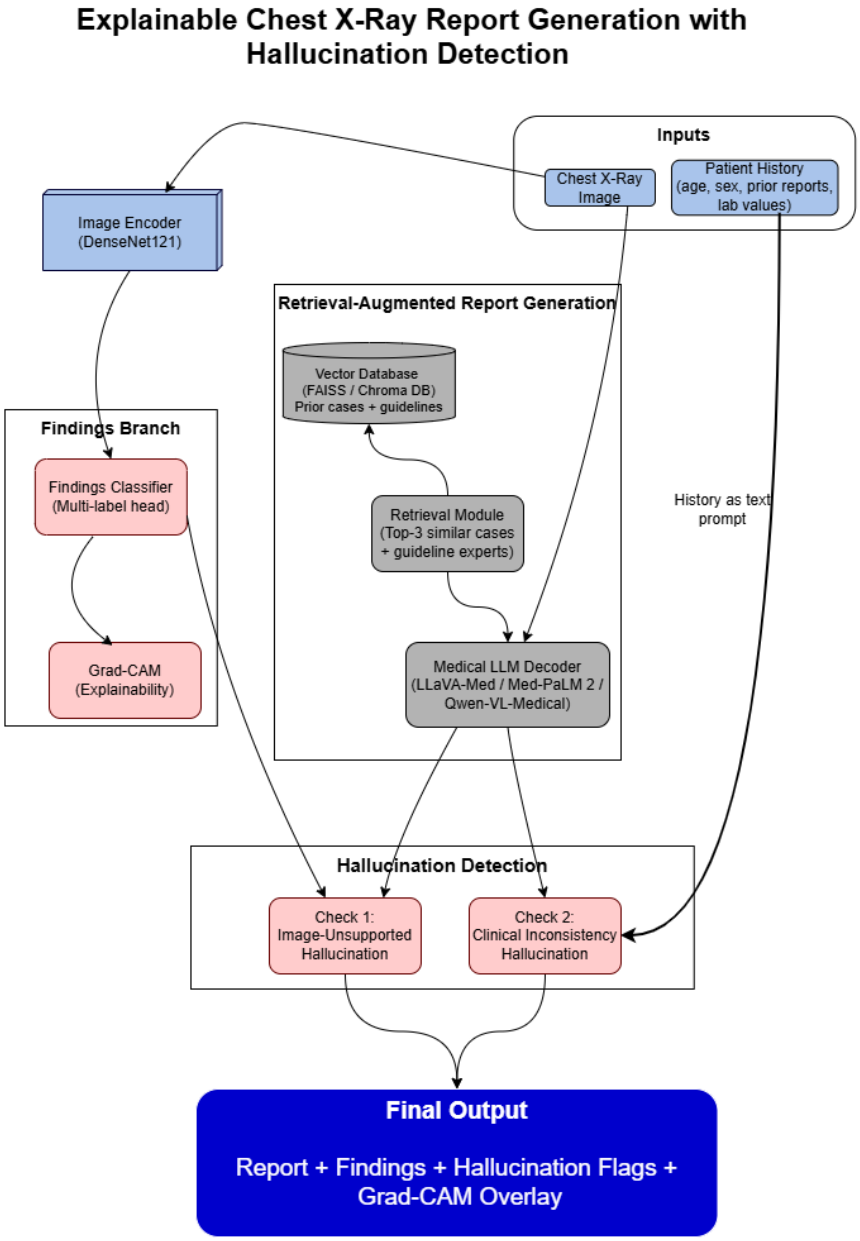


Figure 1: Overall architecture of the proposed explainable chest X-ray report generation system with retrieval augmentation and built-in hallucination detection.

Figure 1 presents the overall architecture of the proposed system. The system takes a chest X-ray image and available patient history as its primary inputs. The chest X-ray is processed by a DenseNet121-based image encoder, whose learned representation is used by the findings branch for multi-label clinical finding classification. Grad-CAM is applied to the findings branch to identify image regions contributing to the predicted findings and

provide visual explanations.

The image representation and patient history are also provided to the retrieval-augmented report generation module. The retrieval module searches a vector database containing relevant prior cases and medical guideline information to retrieve supporting clinical context. The retrieved information, together with the X-ray information and patient history, is provided to the Qwen2-VL-2B vision-language model for generating the radiology report.

The generated report is subsequently processed by the hallucination detection module. Two complementary checks are performed: an image-supported hallucination check, which identifies generated findings that are not supported by the image-based findings classifier, and a clinical-consistency check, which identifies statements that conflict with the available patient history.

The final output combines the generated report, predicted clinical findings, hallucination flags, and Grad-CAM visualization to provide an explainable report-generation result.

6.2 Architecture Components

6.2.1 Input Layer

The system receives two primary inputs: the chest X-ray image and available patient history. The X-ray provides the visual evidence required for finding classification and report generation, while patient history provides additional clinical context.

6.2.2 Image Encoder

DenseNet121 is used as the image encoder. The encoder transforms the preprocessed chest X-ray into a high-level visual representation that is subsequently used by the findings classification and report-generation components.

6.2.3 Findings Branch

The findings branch uses a multi-label classification head to predict the 16 canonical clinical findings. These predictions provide structured image-based evidence that can later be used for hallucination verification.

6.2.4 Grad-CAM Explainability

Grad-CAM is applied to the image-classification branch to produce heatmaps showing regions that contribute to the predicted findings. These heatmaps provide a visual explanation of the model's image-based predictions.

6.2.5 Retrieval Module

The retrieval module uses a vector database containing relevant prior cases and medical guideline information. The module retrieves the most relevant contextual information for the current input.

6.2.6 Medical Vision-Language Model

The retrieved information, patient history, and chest X-ray are supplied to Qwen2-VL-2B for automated radiology report generation.

6.2.7 Hallucination Detection Module

The generated report is checked using two complementary verification mechanisms. The first checks whether generated findings are supported by the image-based classifier, while the second checks whether generated findings are consistent with patient history.

6.2.8 Final Output

The final output consists of the generated radiology report, predicted clinical findings, hallucination flags, and Grad-CAM visualization.

7 Technologies Used

The major technologies used in the proposed system are summarized in Table 3.

Table 3: Technologies Used in the Proposed System

Category	**Technology**
Programming Language	Python
Deep Learning Framework	PyTorch
Image Classification Model	DenseNet121
Explainability	Grad-CAM
Text Embedding Model	all-MiniLM-L6-v2
Vector Database / Search	FAISS
Vision-Language Model	Qwen2-VL-2B
Parameter-Efficient Fine-Tuning	LoRA
Model Quantization	4-bit Quantization
Transformer Framework	Hugging Face Transformers
Data Processing	NumPy, Pandas
Visualization	Matplotlib

8 Methodology

The proposed framework consists of a sequential pipeline for automated chest X-ray analysis and radiology report generation. The methodology comprises image preprocessing, clinical finding label preparation, multi-label pathology classification, class-imbalance handling, visual explanation, retrieval-augmented contextualization, patient-history integration, vision-language report generation, and hallucination verification.

8.1 Image Preprocessing

The raw chest X-ray images undergo a standardized preprocessing pipeline before being supplied to the classification and vision-language components. The preprocessing stage consists of image-view selection, contrast enhancement, resizing, and channel conversion.

8.1.1 Frontal Image Selection

The dataset contains both frontal and lateral chest X-ray projections. Since the proposed system primarily focuses on standard frontal chest radiographs, frontal projections are

selected for the image-analysis pipeline. This results in 3,818 frontal images from the complete collection of 7,466 X-ray images.

8.1.2 Contrast Enhancement Using CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to improve the local contrast of chest X-ray images while limiting excessive amplification of noise.

Unlike global histogram equalization, CLAHE operates on local regions of the image and applies histogram clipping before redistribution. This allows subtle structures in the radiograph to be enhanced while limiting excessive contrast amplification.

8.1.3 Image Resizing

Following contrast enhancement, images are resized to a standardized spatial resolution of 224×224 pixels. Standardized image dimensions ensure compatibility with the DenseNet121-based classification network and provide a consistent input representation.

8.1.4 Channel Conversion

The preprocessed images are converted into three-channel RGB representations so that they are compatible with the pretrained DenseNet121 architecture.

8.2 Clinical Finding Label Preparation

The clinical information associated with each study is transformed into a structured set of 16 canonical chest X-ray findings. The `MeSH` and `Problems` fields provide structured information that can be used during the finding-label extraction process.

Since multiple abnormalities may be present in the same radiograph, each image is represented using a binary multi-label representation containing one binary value for each of the 16 canonical findings.

A value of one indicates the presence of the corresponding finding, whereas zero indicates its absence according to the structured labels generated from the available clinical information.

Multi-label binarization allows a single radiograph to be associated with multiple findings simultaneously and provides a structured target representation for training the DenseNet121 classifier.

8.3 DenseNet121-Based Finding Classification

DenseNet121 is employed as the image-based feature extraction and classification architecture. DenseNet introduces direct connections between each layer and all preceding layers within a dense block.

The dense connectivity mechanism enables feature reuse and improves gradient propagation through the network. This is particularly useful for medical image analysis because both low-level visual structures and higher-level anatomical representations can contribute to identifying abnormalities.

The final representation produced by the DenseNet121 backbone is passed to a multi-label classification layer corresponding to the 16 canonical findings.

8.4 Class-Imbalance Handling

Medical imaging datasets commonly contain an unequal distribution of pathological findings. Some abnormalities occur considerably less frequently than others. Direct optimization on an imbalanced dataset can cause the model to favor common classes and easy negative examples.

To reduce this effect, the classification pipeline incorporates class-balancing strategies and focal loss.

8.4.1 Balanced Sampling

Balanced sampling modifies the probability with which training samples are selected during mini-batch construction. Underrepresented pathological cases are assigned greater sampling importance so that they are represented more frequently during training.

8.4.2 Focal Loss

Focal loss is used to reduce the contribution of easily classified examples and focus optimization on difficult examples.

The loss gives greater importance to difficult examples while reducing the contribution of highly confident and correctly classified samples. A class-balancing component can also be used to compensate for differences in the frequency of positive and negative examples.

8.5 Mixup Data Augmentation

Mixup augmentation is employed during classifier training to improve generalization. Two training examples are combined to produce a synthetic sample and corresponding mixed target.

The mixing coefficient is sampled from a beta distribution, and the same coefficient is used to combine the corresponding labels.

Mixup encourages the classifier to learn smoother decision boundaries and reduces overfitting to individual training examples.

8.6 Visual Explainability Using Grad-CAM

To improve the interpretability of the pathology classifier, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed.

Grad-CAM uses gradients associated with a target clinical finding to determine the importance of different convolutional feature maps. The important feature maps are combined to generate a class-specific heatmap.

The resulting heatmap highlights image regions that contribute strongly to the prediction of the selected finding. These visual explanations can help in understanding whether the classifier is focusing on anatomically relevant regions of the chest radiograph.

8.7 Retrieval-Augmented Generation

To provide additional contextual information to the report-generation model, a retrieval-augmented generation (RAG) module is incorporated using FAISS.

The retrieval component maintains an indexed collection of relevant information, including prior cases and medical guidelines. The input context is converted into an embedding representation, which is compared against the indexed representations to retrieve semantically relevant information.

Semantic similarity is used to compare the input query representation with stored document and case representations.

The documents with the highest similarity values are selected as relevant contextual information.

The retrieved context is then combined with the input X-ray and patient information before being provided to the vision-language model.

8.8 Patient-History Integration

Patient clinical history is incorporated as an additional source of context for report generation and subsequent verification.

The available patient information is converted into a structured textual representation. This information can contain clinically relevant context that may influence the interpretation of radiographic findings.

The combined input to the report-generation model consists conceptually of the X-ray, retrieved information, and patient history. These sources are jointly supplied to the vision-language model during report generation.

8.9 Qwen2-VL-2B Vision-Language Model

Qwen2-VL-2B is employed as the vision-language model for automated radiology report generation. The model receives multimodal information consisting of the chest X-ray together with contextual information generated by the retrieval and patient-history modules.

The model generates the radiology report autoregressively, conditioning each generated token on the preceding tokens, the input image, and the available clinical context.

The generated report is subsequently passed to the verification stage for hallucination analysis.

8.10 Parameter-Efficient Fine-Tuning Using LoRA

To adapt the pretrained Qwen2-VL-2B model to the radiology report-generation task while reducing computational requirements, Low-Rank Adaptation (LoRA) is employed.

LoRA introduces a trainable low-rank update to selected pretrained weight matrices while keeping the original model parameters fixed.

The trainable adaptation matrices have substantially fewer parameters than the original model weights. This reduces the number of trainable parameters compared with full-model fine-tuning.

8.11 4-Bit Quantization

Low-bit quantization is used to reduce the memory requirements of the vision-language model during fine-tuning.

In the proposed implementation, the pretrained model is loaded using 4-bit quantization. This reduces the memory required to store model parameters and enables parameter-efficient fine-tuning using LoRA on limited computational resources.

The combination of 4-bit quantization and LoRA enables adaptation of the pretrained vision-language model while reducing memory consumption.

8.12 Automated Radiology Report Generation

After fine-tuning, the Qwen2-VL-2B model generates a radiology report from an unseen chest X-ray together with the retrieved context and patient history.

The model generates the report sequentially until an end-of-sequence condition is reached or the maximum generation length is exceeded.

The generated report is subsequently passed to the verification stage for hallucination analysis.

8.13 Dual Hallucination Detection

The final component of the proposed framework performs verification of the generated report using two complementary sources of evidence: image-based findings and patient clinical history.

8.13.1 Image-Supported Finding Verification

The first verification stage compares clinical findings extracted from the generated report with the findings predicted by the DenseNet121 classifier.

The findings predicted by the image classifier are compared with the findings extracted from the generated report. A finding appearing in the generated report but not supported by the image-based classifier is flagged for further verification.

8.13.2 Clinical Consistency Verification

The second verification stage compares generated findings against the available patient history.

The generated findings are evaluated for consistency with this contextual information. Findings that conflict with the available patient information are categorized as potential clinical inconsistency hallucinations.

8.13.3 Final Verification Output

The complete hallucination verification output consists of image-supported and clinically inconsistent findings. The two verification results are combined to identify generated statements that require additional review.

8.14 End-to-End Inference Pipeline

During inference, an unseen chest X-ray is first passed through the preprocessing module. The preprocessed image is then supplied to the DenseNet121 classifier to obtain predictions for the 16 canonical findings.

Grad-CAM is subsequently applied to generate visual explanations corresponding to the predicted findings. In parallel, relevant information is retrieved from the FAISS-based knowledge and case index.

Patient history is then combined with the retrieved information and the original X-ray. The resulting multimodal context is provided to the fine-tuned Qwen2-VL-2B model, which generates a draft radiology report.

The draft report is then passed to the hallucination verification module. The final output contains the generated report, predicted findings, hallucination flags, and Grad-CAM visualization.

9 Algorithm / Pseudo Code

9.1 Algorithm: Explainable Chest X-Ray Report Generation with Hallucination Detection

Input: Chest X-ray image I, patient history H

Output: Generated report R, predicted findings F, hallucination flags Z, Grad-CAM visualization G

1. Preprocess the input chest X-ray image.
2. Select the frontal X-ray projection and apply contrast enhancement, resizing, and channel conversion.
3. Pass the preprocessed image through the DenseNet121 image encoder.
4. Use the multi-label classification head to predict the 16 canonical clinical findings.

5. Generate Grad-CAM visualization for the predicted findings.
6. Construct the retrieval query using the available image-related and clinical information.
7. Search the FAISS vector database and retrieve the most relevant prior cases and medical guideline information.
8. Combine the retrieved context with the chest X-ray and available patient history.
9. Provide the multimodal input to the fine-tuned Qwen2-VL-2B model.
10. Generate the draft radiology report.
11. Extract relevant clinical findings from the generated report.
12. Compare the generated findings with the findings supported by the DenseNet121 classifier.
13. Flag findings that are not supported by the image-based classifier as potential image-unsupported hallucinations.
14. Compare the generated findings with the available patient history.
15. Flag findings that conflict with patient history as potential clinical inconsistency hallucinations.
16. Combine the generated report, predicted findings, hallucination flags, and Grad-CAM visualization.
17. Return the final explainable report-generation output.

9.2 End-to-End Processing Summary

The complete proposed workflow can be summarized as:

Chest X-Ray → Preprocessing → DenseNet121 → Clinical Findings → Grad-CAM → RAG + Patient History → Qwen2-VL-2B → Generated Report → Dual Hallucination Detection → Final Explainable Output

10. RESULTS AND DISCUSSION

The proposed system was evaluated at multiple levels, including chest X-ray classification, visual explainability, report generation, retrieval-assisted generation, and hallucination detection. The experiments combine the DenseNet-121 visual classification results with the multimodal report generation and dual hallucination-shield evaluation.

The current primary report-generation implementation is based on Qwen2-VL-2B, while an additional experimental configuration using MedGemma 1.5 4B with LoRA/QLoRA fine-tuning was also evaluated. The latter is reported as an additional experiment and does not replace the primary architecture described in the preceding sections.

10.1 Overall System Performance

The visual classification component uses a DenseNet-121 model pretrained through TorchXRayVision and fine-tuned for multi-label chest X-ray classification. The classifier provides pathology probabilities that are subsequently used as a visual source of truth by the hallucination detection module.

The reported evaluation covers 15 clinical findings on a held-out test set of 578 samples. The classifier achieved a macro-AUROC of 75.42% and a Hamming accuracy of 87.52%. The exact match ratio across the 15 pathology states was 37.72%.

The report generation component was evaluated using standard natural language generation metrics, including BLEU-1, ROUGE-L, METEOR, and CIDEr. In addition, two hallucination shields were evaluated independently and together. Shield 1 verifies generated clinical entities against the DenseNet-121 pathology predictions, while Shield 2 checks temporal and chronicity consistency against patient history.

10.2 DenseNet-121 Classification Results

The DenseNet-121 classifier was evaluated using multiple complementary metrics. The evaluation included per-pathology performance, training convergence, comparison between classification phases, and confusion matrices for the major findings.

10.2.1 Per-Pathology Performance

Figure 2 presents the per-pathology performance of the DenseNet-121 classifier. The results demonstrate that performance varies substantially across the different clinical findings, primarily due to differences in class prevalence and visual characteristics.

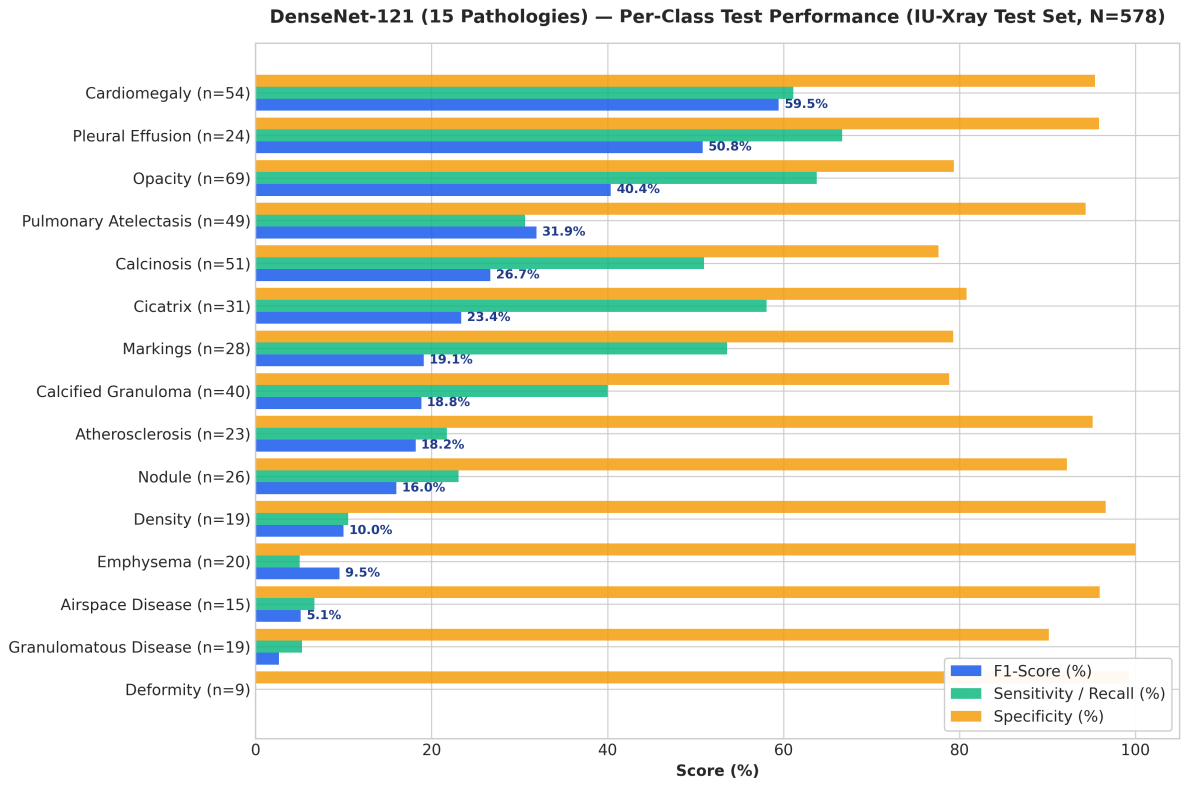


Figure 2: Per-class test performance of the DenseNet-121 classifier across 15 clinical findings.

The strongest F1-score was obtained for cardiomegaly (59.46%), followed by pleural effusion (50.79%) and opacity (40.37%). Several low-prevalence findings produced substantially lower F1-scores, indicating the difficulty of learning rare clinical patterns from the available training data.

The overall macro-AUROC was 75.42%, indicating that the model retains useful ranking capability across the multi-label classification task even when individual thresholded classification performance varies between pathologies.

10.2.2 Training Dynamics

Figure 3 shows the training and validation behavior of the classifier.

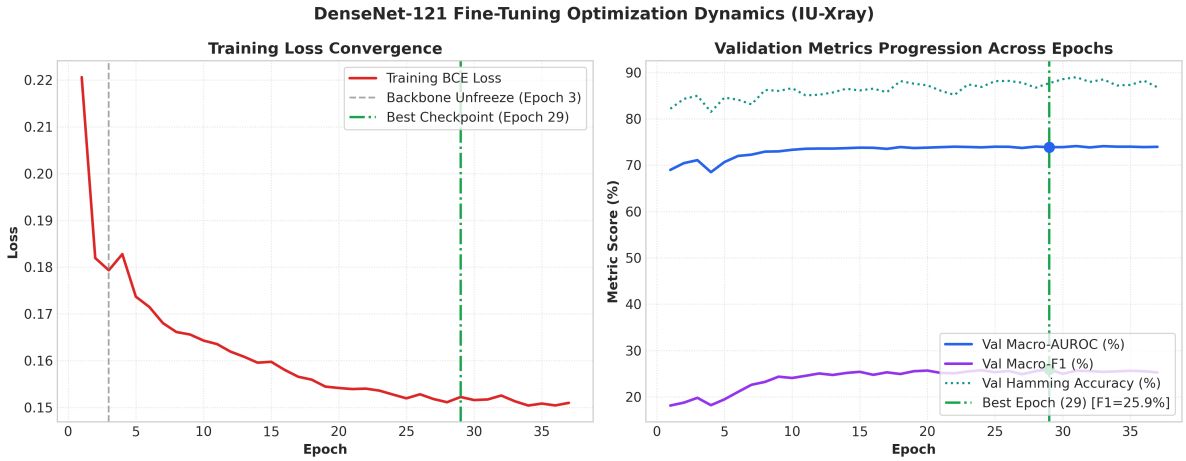


Figure 3: Training dynamics of the DenseNet-121 classifier.

The training curves show progressive reduction in training loss together with changes in validation macro-AUROC and macro-F1. The training procedure used class-imbalance-aware binary cross-entropy and cosine annealing to improve optimization across the multiple pathology labels.

The difference between macro-AUROC and macro-F1 also demonstrates the importance of evaluating both ranking performance and threshold-dependent classification performance. A model may distinguish positive and negative samples reasonably well while still producing lower F1-scores for rare pathologies.

10.2.3 Phase-wise Classification Comparison

The classification development was evaluated in two phases. The comparison between the initial seven-finding configuration and the expanded 15-finding configuration is shown in Figure 4.

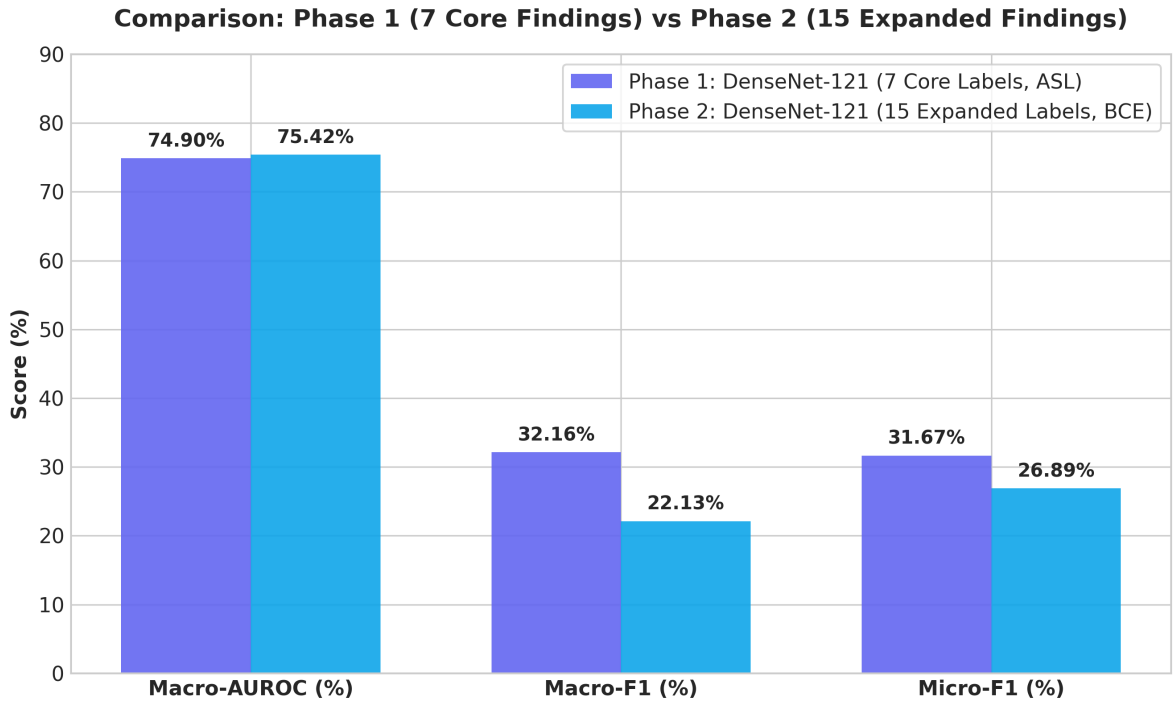


Figure 4: Comparison between the initial seven-finding and expanded 15-finding classification configurations.

The macro-AUROC increased from 74.90% in the seven-finding configuration to 75.42% in the expanded configuration. However, the macro-F1 decreased from 32.16% to 22.13%, while the micro-F1 decreased from 31.67% to 26.89%.

This difference is consistent with the increased difficulty introduced by the additional pathology classes, particularly the rare findings. Thus, the expanded classifier provides broader clinical coverage but presents a more challenging multi-label prediction problem.

10.2.4 Confusion Matrix Analysis

Figure 5 presents confusion matrices for four of the major pathologies.

For cardiomegaly, the classifier achieved an F1-score of 59.46%, with 61.11% sensitivity and 95.42% specificity. Pleural effusion achieved an F1-score of 50.79%, with 66.67% sensitivity and 95.85% specificity.

Opacity achieved an F1-score of 40.37%, with 63.77% sensitivity and 79.37% specificity. Pulmonary atelectasis achieved an F1-score of 31.91%, with 30.61% sensitivity and 94.33% specificity.

The results indicate that the classifier generally maintains high specificity for many findings, while sensitivity and F1-score are more variable, particularly for low-frequency

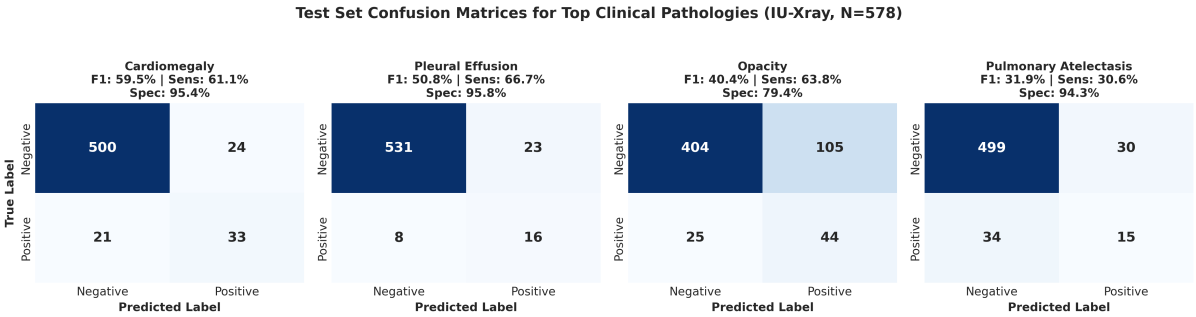


Figure 5: Confusion matrices for selected high-frequency pathologies.

pathologies.

10.3 Detailed Classification Results

Table 4 provides the detailed performance for the 15 evaluated clinical findings.

Table 4: Per-pathology performance of the DenseNet-121 classifier.

Pathology	**Positives**	**Threshold**	**F1 (%)**	**Sens. (%)**	**Spec. (%)**
Cardiomegaly	54	0.36	59.46	61.11	95.42
Pleural Effusion	18	0.22	50.79	66.67	95.85
Opacity	69	0.16	40.37	63.77	79.37
Pulmonary Atelectasis	49	0.28	31.91	30.61	94.33
Cicatrix	23	0.15	26.67	26.09	96.40
Markings	28	0.20	23.38	32.14	92.18
Calcified Granuloma	46	0.25	19.11	32.61	89.66
Calcinosis	48	0.35	18.82	16.67	97.17
Atherosclerosis	23	0.18	18.18	21.74	95.14
Nodule	26	0.05	16.00	23.08	92.21
Density	19	0.07	10.00	10.53	96.60
Emphysema	20	0.33	9.52	5.00	100.00
Airspace Disease	15	0.18	5.13	6.67	95.91
Granulomatous Disease	19	0.05	2.67	5.26	90.16
Deformity	9	0.09	0.00	0.00	99.30
Macro Average	–	0.19	**22.13**	**33.14**	**93.98**

The results show that the model performs considerably better on common findings than on rare pathologies. This is particularly visible in the difference between cardiomegaly and pleural effusion and the low-prevalence classes such as deformity and granulomatous disease.

10.4 Report Generation Results

The report generation component converts the visual and contextual information into structured radiological findings and impressions.

The primary report-generation implementation described in the current system uses Qwen2-VL-2B. In addition, an experimental MedGemma 1.5 4B LoRA/QLoRA configuration was evaluated and is included here for comparison and future migration planning.

Table 5: Comparison of report generation configurations.

Model	BLEU-1	ROUGE-L	METEOR	CIDEr	Val. Loss
Pretrained Zero-shot	11.20%	12.45%	9.80%	0.0410	2.4510
Qwen-VL-Chat LoRA	19.84%	16.35%	14.20%	0.0890	1.2840
MedGemma 1.5 4B LoRA	23.62%	19.02%	17.37%	0.1197	1.0083

The MedGemma experiment used 4-bit QLoRA fine-tuning for three epochs and 1,497 training steps. The LoRA configuration used rank $r = 16$ and $\alpha = 32$, with AdamW optimization and a learning rate of 1×10^{-4}. The reported training loss decreased from approximately 1.84 to 0.57, while the best validation loss was 1.0083.

Compared with the zero-shot configuration, the MedGemma experiment improved BLEU-1 by 12.42 percentage points, ROUGE-L by 6.57 percentage points, METEOR by 7.57 percentage points, and CIDEr by 0.0787. The validation loss decreased by 1.4427.

These natural language generation metrics measure similarity between generated and reference reports. They should therefore be interpreted as language-generation metrics rather than direct measures of clinical factuality or hallucination safety.

The MedGemma configuration is currently treated as an additional experimental result. A complete migration of the primary Qwen2-VL-2B pipeline to MedGemma is planned as future work after the present review.

10.5 Clinical Vector RAG

The retrieval component uses a TF-IDF and cosine-similarity-based vector retrieval mechanism over a curated collection of 308 clinical documents. The retrieval corpus contains relevant ACR guidelines and prior clinical cases.

For each input case, the system retrieves the top three relevant text chunks. These retrieved passages provide additional clinical context to the report-generation stage and help constrain generated statements to information supported by clinical references and previously observed cases.

The retrieval mechanism is therefore used as a contextual grounding component rather

than as a replacement for the image-based classifier.

10.6 Qualitative Case Studies

Case Study 1: Chronic Versus Acute Fracture

A 58-year-old patient had a history of a left clavicle fracture that had occurred in 2024 and had already healed. An unguarded VLM generated a statement suggesting an acute non-displaced fracture.

The visual classifier produced a low probability for an active acute fracture, while the longitudinal shield identified a contradiction between the generated acute statement and the patient's historical information.

The final corrected interpretation described the finding as a remodeled healed fracture without evidence of a new acute injury.

This case demonstrates the importance of combining visual evidence with longitudinal patient history.

Case Study 2: Cardiomegaly and Post-Surgical Findings

In the second case, the reference findings included cardiomegaly, prominent pulmonary arteries, sternotomy wires, and clear lungs.

The classifier detected cardiomegaly, while pleural effusion and consolidation were ruled out. Grad-CAM localization highlighted the cardiac region. The generated report described borderline cardiomegaly, prominent pulmonary vasculature, intact sternal closure wires, and no focal consolidation or pleural effusion.

The visual factuality shield therefore allowed the clinically supported findings to pass the audit.

Case Study 3: Prevention of Unsupported Pleural Effusion

The third case contained basilar atelectasis without confluent consolidation or pleural effusion.

The classifier detected atelectasis while the probability for pleural effusion remained below its decision threshold. The hallucination prevention mechanism consequently prevented an unsupported statement of bilateral pleural effusion from being accepted.

This example demonstrates the role of the visual classifier as a source-of-truth signal for the report auditing stage.

10.7 Discussion

The experimental results demonstrate that the proposed framework addresses multiple aspects of automated chest X-ray interpretation rather than focusing exclusively on report generation.

The DenseNet-121 classifier provides a visual evidence layer with a macro-AUROC of 75.42% across 15 findings. However, the variation in per-pathology F1-score indicates that rare clinical findings remain challenging.

The report-generation experiments show improvements in standard language generation metrics when fine-tuning is applied. The additional MedGemma experiment achieved BLEU-1 of 23.62%, ROUGE-L of 19.02%, METEOR of 17.37%, and CIDEr of 0.1197. These values provide evidence of improved reference-report similarity, although they do not independently establish clinical factuality.

The hallucination-shield results provide a complementary evaluation. Shield 1 achieved a reported audit accuracy of 95.2%, while Shield 2 detected 25 out of 25 injected contradictions in its reported benchmark. The combined system achieved a reported audit accuracy of 96.8%.

The ablation study further indicates that the combination of visual evidence, longitudinal patient history, and retrieval-based context can improve the reported clinical acceptability of generated reports.

Overall, the current system demonstrates the feasibility of combining multimodal report generation with explicit post-generation factuality checks. Further evaluation on larger and clinically diverse datasets is required before the system can be considered for real-world clinical use.

11. CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

This project presents RadVision AI, a multimodal chest X-ray analysis framework designed to combine visual classification, explainability, retrieval-assisted generation, and hallucination detection.

The system uses a DenseNet-121-based multi-label classifier to obtain visual pathology probabilities and Grad-CAM to provide visual interpretability. A clinical vector retrieval module provides additional context from ACR guidelines and prior cases. The report-generation stage uses Qwen2-VL-2B as the current primary multimodal generation con-

figuration.

The evaluation of the visual classifier across 15 findings produced a macro-AUROC of 75.42%, Hamming accuracy of 87.52%, and exact match ratio of 37.72% on the reported 578 held-out samples.

The system also incorporates a dual hallucination shield. Shield 1 checks generated clinical entities against image-based classifier predictions, while Shield 2 verifies temporal and chronicity consistency with patient history. The reported Shield 1 evaluation achieved 95.2% audit accuracy, while Shield 2 detected 25 out of 25 injected contradictions in its specific benchmark. The unified evaluation reported 96.8% accuracy.

An additional experiment using MedGemma 1.5 4B with LoRA/QLoRA fine-tuning achieved BLEU-1 of 23.62%, ROUGE-L of 19.02%, METEOR of 17.37%, and CIDEr of 0.1197. This experiment demonstrates the potential of a medical-domain multimodal model for future development of the report generation component.

The overall results demonstrate the feasibility of integrating visual evidence, clinical context, explainability, and explicit hallucination auditing into a unified chest X-ray analysis workflow.

11.2 Future Scope

Several directions can be explored to improve the proposed system.

1. **Migration to MedGemma:** The complete report-generation pipeline can be migrated from the current Qwen2-VL-2B configuration to the MedGemma 1.5 4B architecture. The migration can include unified fine-tuning, inference, RAG integration, and hallucination evaluation.

2. **Larger and More Diverse Datasets:** Future experiments can incorporate larger multi-institutional chest X-ray datasets containing a broader range of patient populations, acquisition settings, and pathological findings.

3. **Improved Rare-Pathology Detection:** The relatively low F1-scores observed for several rare findings indicate the need for improved class-balancing strategies, targeted augmentation, focal or asymmetric losses, and additional clinical samples.

4. **Advanced Hallucination Benchmarks:** A larger benchmark containing systematically generated and clinically validated hallucination cases can be developed to evaluate the dual shield under a wider variety of failure conditions.

5. **Clinical Expert Evaluation:** Future work should include evaluation by qualified

radiologists to assess factual correctness, clinical relevance, completeness, readability, and usefulness of generated reports.

6. **Improved Longitudinal Reasoning:** The current heuristic chronicity and chronology module can be extended using dedicated temporal reasoning and medical natural language inference models.

7. **Fine-Grained Factual Verification:** Instead of checking only pathology-level entities, future versions can verify attributes such as location, laterality, severity, size, temporal status, and uncertainty.

8. **Enhanced Retrieval-Augmented Generation:** The current TF-IDF and cosine-similarity retrieval mechanism can be extended using dense biomedical embeddings, hybrid retrieval, and reranking models.

9. **Improved Visual Grounding:** Future versions can investigate region-level grounding and more detailed alignment between generated statements and image regions.

10. **Comprehensive Report Evaluation:** In addition to BLEU, ROUGE, METEOR, and CIDEr, future evaluations can incorporate clinically oriented factuality metrics and expert-based assessment.

11. **Efficiency and Deployment:** Quantization, model compression, optimized inference pipelines, and GPU/edge deployment can be investigated to reduce latency and memory requirements.

12. **Prospective Clinical Validation:** Before practical clinical deployment, the system should undergo prospective validation under appropriate clinical, ethical, privacy, and regulatory requirements.

The long-term objective is to develop a reliable multimodal radiology assistant in which generated reports are not only linguistically coherent but are also grounded in visual evidence, patient history, and trusted clinical knowledge.

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